

Assignment Briefing

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Assignment briefing

Problem setup:

s1	s2	s3	s4	s5
s6	s7			s8
s9		s10	s11	s12

- 1 target area, 3 forbidden areas
- 12 states, each of which has 5 actions
- The reward setting is $r_{\text{boundary}} = r_{\text{forbidden}} = -1$, and $r_{\text{target}} = 1$.
- The discount rate is $\gamma = 0.9$.

Assignment briefing

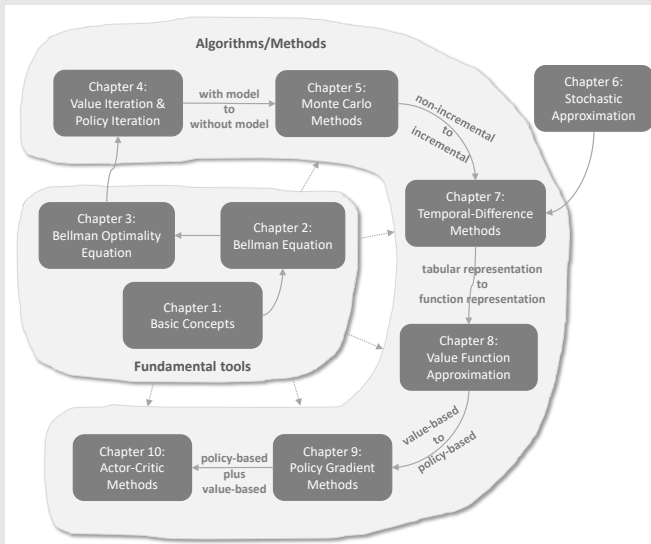
Pay attention to the following to avoid unnecessary mistakes:

- You **should** use the grid-world task specified in the figure. You **should not** use other tasks.
- The forbidden areas are **not accessible** and do **not** correspond to states. If the agent attempts to enter, it will get bounced back to its previous state.
- You **should** use the reward and discount rate specified before. You **should not** use other values.
- Each state has 5 actions. You **should** use algorithms to learn which actions are good or bad. You **should not** manually eliminate some actions even though you think they are not good.
- You **should** fully utilize the code I provide for, for example, plotting arrows or trajectories. It will save you time. You **should not** do the plotting from scratch by yourself.

Five assignments:

- A report for each assignment.
- Each report is worth 20% of the final mark.
- Each assignment is the foundation of the next. So make sure to complete each assignment carefully.
- Each assignment takes about ? hours.

Assignment briefing



Assignment 1

▷ Purpose of this assignment:

- Build up the simulation environment.
- Understand the basic concepts in RL.

▷ The code about plotting is available! See the GitHub homepage.

This assignment is important. The code developed in this assignment will be frequently used in the following assignments.

Assignment 1

- ▷ In your report, you should clearly include the following contents.
 - Describe the software that you use. (5% mark)
 - Describe your the reward setting and discount rate. (5% mark)
 - Design any two policies that you like. One should be deterministic and the other should be stochastic. Do the following for each policy.
 - Describe the policy using an array or table. (10%*2 mark)
 - Plot the policy: show arrows at each state to indicate the policy. (10%*2 mark)
 - Specify an arbitrary starting state that you like. Show the trajectory from the starting state following the policy. The length of the trajectory is 50 steps. (10%*2 mark)
 - Calculate and show the discounted return of this trajectory. (10%*2 mark)
- The above plot and calculation should be done automatically by your code. Some key parts of your code should be given and explained concisely.
- Discuss what you have observed about the two policies. (10% mark)